

A Comparative Life Cycle Cost Analysis of Sustainable and Conventional Buildings

Simple Study Guide

Understand the project from zero: LCC, discount rate, energy escalation, sensitivity analysis, methodology, research gap, break-even, and viva preparation.

40 years Baseline lifespan	5% Discount rate	3% Energy escalation
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Prepared from: EPP_LS.pdf, Project_Proposal_EPP.pdf, and EPP_REFLECTIONS.pdf

How to use this guide

Read Sections 1-6 first. They explain the whole project in plain language. Then use Sections 7-13 for methodology and research structure. Finish with the calculation walkthrough and viva-ready summary.

The one sentence to remember

This project tests whether paying more upfront for a sustainable building can lead to lower total cost over the building's life because of lower energy, water, maintenance, and replacement costs.

Part	What you learn
1-3	What the project is about and what Life Cycle Cost means
4-6	Why sustainable buildings may cost more initially but save later
7-9	Discount rate, energy escalation, and sensitivity analysis
10-13	Research questions, data, methodology, break-even, timeline, risks
14-15	How the three documents connect and how to calculate LCC
Final	Viva-ready summary and revision checklist

1. What is the whole project about?

Imagine two similar buildings in Sydney.

Building A: Conventional	Building B: Sustainable
Usually cheaper to construct at the beginning	May cost more initially
May use more electricity and water	Designed to use less energy and water
May have higher operating and maintenance costs	May reduce operating and maintenance costs
Looks cheaper if you compare only Day 1 cost	May become cheaper when you compare many years

Main research question

Even though a sustainable building can cost more today, can it be cheaper when we look at the total cost over its whole life?

The literature used in the project reports approximate ranges such as:

- 5-15% higher initial construction cost for sustainable buildings.
- 20-40% lower energy use.
- 30-50% lower water use.
- 10-25% lower maintenance costs.
- Around 10-20% lower total life-cycle cost in some studies over long analysis periods.

These ranges are background evidence from the literature. They are not automatically the final result of your own Sydney model.

2. What does Life Cycle Cost (LCC) mean?

Life Cycle Cost means all important costs of an asset over its whole life - not only the price paid at the beginning.

Simple definition

$LCC = \text{Initial Cost} + \text{Operating Costs} + \text{Maintenance} + \text{Replacements} + \text{Disposal}$, with future costs converted to present value.

Car example

Car A costs \$30,000 to buy and Car B costs \$35,000. If you look only at purchase price, Car A looks cheaper. But after 10 years, fuel, servicing, repairs, tyres, and other costs could make Car A cost \$80,000 in total while Car B costs \$65,000. Then Car B was cheaper over its life.

The same idea applies to buildings. A building that is expensive on Day 1 may still be economical over 30-50 years.

3. What exactly are you comparing?

Cost category	Sustainable building	Conventional building
Initial construction	Usually higher	Base / lower
Energy	Lower	Higher
Water	Lower	Higher
Maintenance	Potentially lower	Moderate / higher
Replacement	Depends on systems and materials	Depends on systems and materials
Disposal	Included in full LCC model	Included in full LCC model
Total long-term LCC	To be calculated	To be calculated

Important: your project is not asking which building is cheaper to construct. It is asking which building is cheaper to own and operate over decades.

4. Why can the sustainable building cost more initially?

Sustainable buildings may use better-performing systems and materials: improved insulation, efficient lighting, high-efficiency HVAC, water-saving fixtures, controls, renewable-energy systems, or other design measures.

Simple initial-cost example

Item	Conventional	Sustainable
Base construction cost	\$10,000,000	\$10,000,000
Sustainability premium	\$0	+10% = \$1,000,000
Initial cost	\$10,000,000	\$11,000,000

At first, the sustainable building is \$1 million more expensive. The whole project asks whether future savings can recover this extra amount.

5. Where do the long-term savings come from?

Energy savings

Suppose the conventional building spends \$200,000 per year on energy. If the sustainable building uses 30% less energy:

Example

$30\% \times \$200,000 = \$60,000$ saved per year. Sustainable energy cost = \$140,000 per year.

Ignoring discounting for a moment, $\$60,000 \times 40 \text{ years} = \2.4 million of cumulative energy savings.

Water savings

If the conventional building spends \$50,000 per year on water and the sustainable building saves 40%, the saving is \$20,000 per year and the sustainable water cost is \$30,000 per year.

Maintenance savings

If conventional maintenance costs \$100,000 per year and the sustainable building saves 20%, the saving is \$20,000 per year.

What the project is really testing

Is the extra money paid today smaller than the present value of the money saved over the next 30-50 years?

6. Why does the project use a 40-year baseline?

Buildings are long-life assets. The project uses 40 years as a central baseline and then checks 30-year and 50-year cases in sensitivity analysis.

Scenario	Purpose
30 years	Shorter analysis period
40 years	Baseline case
50 years	Longer analysis period

A longer analysis period gives operating savings more time to accumulate, but the financial value of distant costs and savings is also affected by discounting.

7. What is a discount rate?

A dollar today is normally treated as more valuable than the same dollar many years in the future because money available today can be invested or used immediately. Discounting converts future costs into an equivalent value today.

Baseline

The project uses a 5% discount rate as the main case, then tests 4%, 5%, 6%, and 7%.

One-year example

If a cost of \$100 occurs one year from now and the discount rate is 5%:

$$\text{Present Value} = \text{Future Cost} / (1 + \text{discount rate})^{\text{year}}$$

$$\$100 / 1.05 = \$95.24 \text{ today.}$$

So under a 5% discount rate, paying \$100 next year is economically similar to paying about \$95.24 today.

Why test several discount rates?

Because the result can change depending on how strongly future costs are discounted. A higher discount rate usually makes distant future savings less valuable in present-value terms. Testing several rates shows whether the conclusion depends heavily on one financial assumption.

8. What is energy-price escalation?

Energy-price escalation means assuming electricity or other energy prices change over time instead of staying constant forever.

Baseline

The project uses 3% annual energy-price escalation, then tests 2%, 3%, 4%, and 5%.

Year	Energy cost if Year 1 is \$100,000 and escalation is 3%
1	\$100,000
2	\$103,000
3	\$106,090
4	\$109,273

Energy escalation matters because the sustainable building uses less energy. If energy prices rise faster, the dollar value of energy savings can become more important.

9. What is sensitivity analysis?

Sensitivity analysis means changing important assumptions and recalculating the model to see whether the conclusion changes.

Plain-English meaning

Change the assumptions, then check what happens to the answer.

Variable	Baseline	Sensitivity cases
Discount rate	5%	4%, 5%, 6%, 7%
Energy-price escalation	3%	2%, 3%, 4%, 5%
Building life	40 years	30, 40, 50 years

Example: if the baseline model says the sustainable LCC is \$20 million and the conventional LCC is \$23 million, sensitivity analysis asks whether that difference still exists when the discount rate, energy escalation, or lifespan changes.

Memory trick: 40 - 5 - 3

40 years lifespan - 5% discount rate - 3% energy escalation.

10. What is the research gap?

A research gap is the part of a topic that existing studies have not answered well enough. Your project identifies three important gaps:

- Limited Sydney-specific data. Many studies use global or broad Australian averages.
- Too few studies combine sustainable-building performance with realistic Quantity Surveying cost analysis.

- Different studies use different discount rates, energy assumptions, and lifespans, making direct comparison difficult.

Why your project exists

Existing research suggests sustainable buildings can save money, but your project aims to compare them under a consistent LCC framework with a Sydney focus.

11. Research question and sub-questions

Primary research question

Primary question

How do sustainable and conventional buildings compare in terms of life-cycle cost performance under Sydney construction and operating conditions?

Sub-questions in simple words

1. How much operating cost can sustainable buildings save in energy, water, and maintenance?
2. What happens to the result when financial assumptions change?
3. What is the percentage difference between the final LCCs?
4. Which cost category creates the biggest saving?
5. How can LCC help developers, quantity surveyors, and decision-makers plan better buildings?

12. Where will the data come from?

The project mainly uses secondary data - information that already exists and was collected or published by other people or organisations.

- Peer-reviewed academic publications.
- Infrastructure Australia.
- Green Building Council of Australia (GBCA).
- Australian Bureau of Statistics (ABS).

You are not physically constructing two buildings and monitoring them for 40 years. Instead, you build comparable financial models using published evidence and consistent assumptions.

13. Methodology: the four phases

Phase	What happens	Main output
Phase 1	Collect literature, cost data, energy/water/maintenance evidence, and define assumptions.	Literature summary + data sources
Phase 2	Build a conventional-building LCC model and a sustainable-building LCC model.	Comparative cost tables
Phase 3	Change discount rate, energy escalation, and lifespan and recalculate.	Sensitivity graphs/tables
Phase 4	Compare results, calculate percentage differences and break-even, explain limitations, and make recommendations.	Final report + recommendations

14. What is the break-even point?

Break-even is the point where the extra upfront cost of the sustainable building has been recovered by cumulative operating savings.

Starting point	Example
Conventional initial cost	\$10,000,000
Sustainable initial cost	\$11,000,000
Extra upfront cost	\$1,000,000
Annual operating saving	\$150,000

As yearly savings accumulate, the sustainable option gradually closes the initial \$1 million gap. The year in which the discounted cumulative savings cover that gap is the break-even year.

15. Why Sydney?

Construction economics are location-specific. Sydney can have different labour costs, material prices, energy prices, regulations, climate conditions, and market conditions from other cities or countries. That is why using Sydney-relevant evidence improves the usefulness of the analysis.

16. Project timeline

Weeks	Stage	Output
1-3	Literature review and data collection	Literature summary and data sources
4-6	Build LCC models	Comparative cost tables
7-9	Sensitivity analysis	Sensitivity-analysis graphs
10-12	Comparative evaluation and report writing	Final report and recommendations

17. Main project risks and responses

Risk	Response
Limited Sydney-specific data	Use several Australian government, industry, and academic sources; clearly state limitations.
Different studies give different results	Use a consistent set of modelling assumptions.
Financial calculation mistakes	Cross-check formulas and totals.
Time delays	Use staged milestones and a 12-week workflow.
Future energy prices are uncertain	Use sensitivity analysis rather than relying on one forecast.

18. Ethics and Indigenous sustainability

Ethics

- Use and cite secondary sources correctly.
- Maintain academic integrity.
- Record financial assumptions and calculations transparently.
- No private human-participant data is required for the proposed study.

Indigenous sustainability

The proposal acknowledges that Indigenous Australian perspectives often emphasise long-term stewardship of land and resources, which can align conceptually with sustainable-building principles. However, the study does not directly use Indigenous cultural knowledge or materials as research data.

19. How the three documents connect

Document	Main purpose	Simple question it answers
EPP_LS.pdf	Literature review	What do other researchers already know?
Project_Proposal_EPP.pdf	Research plan	What is missing, and how will I investigate it?
EPP_REFLECTIONS.pdf	Learning reflection	What feedback/problems did I face, how did I improve, and what did I learn?

The logic is: previous research -> research gap -> research questions -> methodology -> analysis -> reflection.

20. LCC calculation walkthrough

The mathematical part becomes much easier when you separate it into yearly steps.

Step	What you do
1	Enter initial construction cost for each building.
2	Estimate Year 1 energy, water, and maintenance costs.
3	Apply the sustainable saving percentages to create sustainable operating costs.
4	Escalate future energy costs using the chosen annual escalation rate.
5	Discount each future cost back to present value.
6	Add replacement and disposal costs at the appropriate years.
7	Sum all present values to get total LCC for each building.
8	Compare LCCs and calculate the percentage difference.

Core present-value formula

PV of a cost in Year t = Future Cost in Year t / $(1 + r)^t$, where r is the discount rate.

Percentage saving formula

$(\text{Conventional LCC} - \text{Sustainable LCC}) / \text{Conventional LCC} \times 100$

Mini numerical example

Item	Conventional	Sustainable
Initial cost	\$10.0m	\$11.0m
Annual energy cost before escalation	\$200,000	\$140,000
Annual water cost	\$50,000	\$30,000
Annual maintenance cost	\$100,000	\$80,000

In the real model, you would project these costs year by year, escalate the relevant costs, discount each future cost, add replacements/disposal, and then sum the present values.

Example of final comparison

Suppose the final present-value LCCs are: Conventional = \$25 million; Sustainable = \$21 million.

Difference

$\$25\text{m} - \$21\text{m} = \$4\text{m}$ lower LCC for the sustainable case.

Percentage difference

$\$4\text{m} / \$25\text{m} \times 100 = 16\%$.

Correct academic wording: "Under the stated assumptions, the sustainable case has a 16% lower modelled LCC." Avoid claiming this is a universal result.

21. Viva-ready explanation

60-second answer

My project compares sustainable and conventional buildings in Sydney using life-cycle costing. Sustainable buildings can have higher initial costs, but they may reduce energy, water, and maintenance costs during operation. Instead of comparing only construction cost, I compare total cost over the building's life. My baseline uses a 40-year lifespan, 5% discount rate, and 3% annual energy-price escalation. Because future costs are uncertain, I also perform sensitivity analysis using different discount rates, energy-price growth rates, and building lifespans. The aim is to determine whether the higher initial investment can be recovered through long-term savings and how sensitive that result is to the financial assumptions.

22. The most important things to memorise

Topic	Remember this
Problem	Construction cost alone does not show the true long-term cost of a building.
Comparison	Sustainable vs conventional buildings.
Method	Life Cycle Costing (LCC).
Location	Sydney-focused comparison.
Baseline	40 years + 5% discount + 3% energy escalation.
Data	Secondary academic, government, and industry sources.
Costs	Initial + energy + water + maintenance/replacement + disposal.
Uncertainty	Future financial assumptions and limited local data.
Solution	Sensitivity analysis.
Outputs	LCC difference, percentage difference, cost contribution, and break-even point.

23. One important academic caution

Do not overclaim

The percentages such as 5-15%, 20-40%, 30-50%, 10-25%, and 10-20% are literature-derived ranges or modelling inputs. They are not automatically the final result of your own Sydney LCC analysis.

A safer statement is: "The reviewed literature indicates potential long-term savings, and my analysis tests the comparison under consistent assumptions with a Sydney focus."

24. Final revision checklist

- ☐ I can explain LCC without reading the formula.
- ☐ I know why a sustainable building can cost more initially.
- ☐ I can explain discount rate in simple language.
- ☐ I can explain energy-price escalation.
- ☐ I know the baseline: 40 years, 5%, 3%.
- ☐ I can explain sensitivity analysis and why it is needed.
- ☐ I can explain the research gap and why Sydney matters.
- ☐ I know the four methodology phases.
- ☐ I can explain break-even point.
- ☐ I know the difference between literature results and my final model results.
- ☐ I can describe how the literature review, proposal, and reflection connect.

End of study guide | Source documents: EPP_LS.pdf, Project_Proposal_EPP.pdf, EPP_REFLECTIONS.pdf